

- 0 Quick reference
- Conditional moments**
- Tower:** $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$.
 - Total variance:** $\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid Y)] + \text{Var}(\mathbb{E}[X \mid Y])$.
 - Jensen:** f convex $\Rightarrow f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$.
- Gamma / Beta identities**

$$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt, \quad \Gamma(n) = (n-1)!$$

$$\Gamma(\alpha+1) = \alpha \Gamma(\alpha).$$

$$\int_0^\infty t^{\alpha-1} e^{-\beta t} dt = \frac{\Gamma(\alpha)}{\beta^\alpha} \quad (\beta > 0).$$

$$B(\alpha, \beta) = \int_0^1 t^{\alpha-1} (1-t)^{\beta-1} dt = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$$

Change of variables

$$Y = g(X), \text{ } g \text{ smooth invertible: } f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dg^{-1}}{dy} \right|.$$

Linear: $Y = aX \Rightarrow f_Y(y) = \frac{1}{|a|} f_X(y/a)$. **Reciprocal:** $Y = 1/X \Rightarrow f_Y(y) = \frac{1}{y^2} f_X(1/y)$.

Sum of squares

$$\sum_{i=1}^n (x_i - \mu)^2 = \sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2.$$

Use: splits a Normal log-likelihood into a μ -free SSR and a quadratic in μ – feeds directly into the Normal / Gamma-Normal posteriors.

Completing the square

$$a\mu^2 - 2b\mu + c = a\left(\mu - \frac{b}{a}\right)^2 + \left(c - \frac{b^2}{a}\right).$$

Use: a Normal kernel $\exp(-\frac{1}{2}(a\mu^2 - 2b\mu))$ is $\mathcal{N}(b/a, 1/a)$ in μ ; the leftover $c - b^2/a$ is the marginal-likelihood factor.

Distributions			
Family	$f(x) \propto$	$\mathbb{E}[X]; \text{Var}(X)$	Conj.
Beta (α, β)	$x^{\alpha-1}(1-x)^{\beta-1}$	$\frac{\alpha}{\alpha+\beta}, \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	Beta-lhoods
Gamma $(\alpha, 1/\beta)$	$x^{\alpha-1}e^{-\beta x}$	$\alpha/\beta; \alpha/\beta^2$	Pois/Exp/Gam
InvGam (α, β)	$x^{-\alpha-1}e^{-\beta/x}$	$\frac{\beta}{\alpha-1}, \frac{\beta^2}{(\alpha-1)^2(\alpha-2)}$	Normal
Exp (λ)	$\lambda e^{-\lambda x}$	$1/\lambda; 1/\lambda^2$	Gamma
Pareto (α, x_0)	$x^{-\alpha-1}$	$\frac{\alpha x_0}{\alpha-1}, \frac{\alpha x_0^2}{(\alpha-1)^2(\alpha-2)}$	Unif
Unif $(0, \theta)$	$\theta^{-1}\mathbf{1}\{x < \theta\}$	$\theta/2; \theta^2/12$	Pareto
Normal $(\mu, 1/\tau)$	$e^{-\frac{\tau}{2}(x-\mu)^2}$	$\mu; 1/\tau$	Normal
Student-t ν	$(1+x^2/\nu)^{-(\nu+1)/2}$	$0; \nu/(\nu-2)$	–
Bern (p)	$p^x(1-p)^{1-x}$	$p; p(1-p)$	Beta
Bin (n, p)	$\binom{n}{x} p^x (1-p)^{n-x}$	$np; np(1-p)$	Beta
Geom (θ)	$(1-\theta)^x \theta$	$\frac{1-\theta}{\theta}; \frac{1-\theta^2}{\theta^2}$	Beta
NegBin (r, θ)	$\binom{x+r-1}{x} (1-\theta)^x \theta^r$	$\frac{r(1-\theta)}{\theta}; \frac{r(1-\theta)}{\theta^2}$	Beta
Pois (λ)	$\lambda^x / x!$	$\lambda; \lambda$	Gamma
Dir (α)	$\prod_i x_i^{\alpha_i-1}$	$\frac{\alpha_i}{\alpha}, \frac{\alpha_i(\alpha-\alpha_i)}{\alpha^2(\alpha+1)}$	Mult
Mult $(n, \mathbf{p})$	$\prod_i p_i^x / x_i!$	$np_i; np_i(1-p_i)$	Dir

- Notes:**
- Dir marginals: $X_i \sim \text{Beta}(\alpha_i, \alpha - \alpha_i)$ with $\alpha = \sum_j \alpha_j$.
 - Mult cross-moments: $\text{Cov}(X_i, X_j) = -np_i p_j$ ($i \neq j$).
 - Geom / NegBin convention here: failures before r th success ($r = 1$ for Geom).
 - Conj. partner is the *other* family in the conjugate pairing – read row-by-row whether the family acts as prior or likelihood.

- 1 Foundations**
- Bayesian vs frequentist**
- Frequentist:** θ fixed; MLE $\hat{\theta}$, CI from sampling distribution of $\hat{\theta}$.
 - Bayesian:** θ random; prior $\pi(\theta) \rightarrow$ posterior $\pi(\theta \mid x)$.
 - Posterior mean** $= \mathbb{E}[\theta \mid x]$; **posterior variance** $= \text{Var}(\theta \mid x)$.
 - $(1 - \alpha)$ credible set** C : $\mathbb{P}(\theta \in C \mid x) = 1 - \alpha$.
 - HDR** = smallest-volume $C = \{\theta : \pi(\theta \mid x) \geq k_\alpha\}$.
Equal-tail = HDR iff posterior is symmetric unimodal.

- Point estimates by loss**
- Squared loss** $(\hat{\theta} - \theta)^2$: $\hat{\theta} = \mathbb{E}[\theta \mid x]$ (posterior **mean**).
 - Absolute loss** $|\hat{\theta} - \theta|$: $\hat{\theta}$ = posterior **median**.
 - 0-1 loss** $\mathbf{1}\{\hat{\theta} \neq \theta\}$ (or $\rightarrow 0$ width): $\hat{\theta} = \arg \max_\theta \pi(\theta \mid x)$ (**MAP** / posterior **mode**).

Bayes' theorem

$$\pi(\theta \mid x) = \frac{\pi(\theta) L(\theta \mid x)}{m(x)} \propto \pi(\theta) L(\theta \mid x),$$

$$m(x) = \int \pi(\theta) L(\theta \mid x) d\theta.$$

Sequential updating

For observations $x_1, \dots, x_k$,

$$\pi(\theta \mid x_{1:k}) \propto \pi(\theta \mid x_{1:k-1}) \pi(x_k \mid \theta, x_{1:k-1}).$$

Dirichlet via conditional Betas

$(X_1, \dots, X_k) \sim \text{Dir}(\alpha_1, \dots, \alpha_k)$ iff

$$X_1 \sim \text{Beta}(\alpha_1, \alpha_{2+}),$$

$$\frac{X_{i+1}}{1-X_1-\dots-X_i} \mid X_{1:i} \sim \text{Beta}(\alpha_{i+1}, \alpha_{(i+2)+}),$$

with $\alpha_{j+} = \sum_{l \geq j} \alpha_l$. **Use** to simulate Dirichlet by chained Betas.

Multinomial conditional

$(X_1, \dots, X_k) \sim \text{Mult}(n, p_1, \dots, p_k)$:

$$X_{k-1} \mid X_{1:k-2} \sim \text{Bin}(n^*, \tilde{p}),$$

$$n^* = n - \sum_{i=1}^{k-2} x_i, \quad \tilde{p} = \frac{p_{k-1}}{1 - \sum_{i=1}^{k-2} p_i}.$$

Improper / flat priors

$\pi(\theta) \propto 1$ or $\pi(\theta) \propto 1/\theta$ integrates to ∞ , yet the posterior can still be proper.

- Trick:** deflate prior influence to recover frequentist-like answers (e.g. flat prior on μ gives $\bar{x} \pm$ usual SE).
- Must check:** $\int \pi(\theta \mid x) d\theta < \infty$ – improper prior $\nrightarrow$ proper posterior automatically.

Example: Laplace's rule. $\theta \sim \text{Beta}(1, 1)$, $x \sim \text{Bin}(n, \theta)$: posterior $\text{Beta}(1+x, 1+n-x)$, mean $(1+x)/(n+2)$.

2 Discrete priors & Normal inference

Discrete prior

$\theta \in \{\theta_1, \dots, \theta_K\}$ with prior weights π_k . Posterior:

$$\pi(\theta_k \mid x) = \frac{\pi_k L(\theta_k \mid x)}{\sum_j \pi_j L(\theta_j \mid x)}.$$

Pure re-weighting of the PMF; no integral.

Normal mean (precision r known)

$x_i \mid \mu \sim N(\mu, 1/r)$, prior $\mu \sim N(m, 1/t)$:

$$\mu \mid x \sim N(m_n, 1/t_n), \quad t_n = t + nr,$$

$$m_n = \frac{tm + nr\bar{x}}{t_n}.$$

Flat prior: $t \rightarrow 0$, so $\mu \mid x \sim N(\bar{x}, 1/(nr))$ – matches MLE.

Normal precision (mean μ known)

$x_i \mid \tau \sim N(\mu, 1/\tau)$, prior $\tau \sim \text{Gamma}(\alpha, 1/\beta)$:

$$\tau \mid x \sim \text{Gamma}\left(\alpha + \frac{n}{2}, \frac{1}{\beta + \frac{1}{2} \sum_i (x_i - \mu)^2}\right).$$

Gamma-Normal $(\alpha, 1/\beta; m, 1/t)$ (both unknown)

$x_i \mid \mu, \tau \sim N(\mu, 1/\tau)$. Prior: $\tau \sim \text{Gamma}(\alpha, 1/\beta)$ and $\mu \mid \tau \sim N(m, 1/(t\tau))$.

$$\alpha_n = \alpha + \frac{n}{2}, \quad t_n = t + n, \quad m_n = \frac{tm + n\bar{x}}{t_n},$$

$$\beta_n = \beta + \frac{1}{2} \left[\sum_i (x_i - \bar{x})^2 + \frac{(m - \bar{x})^2}{t^{-1} + n^{-1}} \right].$$

Gamma-Normal marginals (Student- t)

$$\mu \mid x \sim m_n + \sqrt{\frac{\beta_n}{\alpha_n t_n}} t_{2\alpha_n},$$

$$X_{n+1} \mid x \sim m_n + \sqrt{\frac{\beta_n(1+t_n)}{\alpha_n t_n}} t_{2\alpha_n}.$$

Same centre, wider scale for the predictive: extra $(1+t_n)$ factor accounts for a new-observation noise on top of posterior uncertainty in μ . **Flat Gamma-Normal.** $t \rightarrow 0$, $\alpha \rightarrow -1/2$, $\beta \rightarrow 0$: $\mu \mid x \sim \bar{x} + (s/\sqrt{n}) t_{n-1}$ with $s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$ – classical one-sample t interval.

3 Conjugate families

Definition

A family $\mathcal{F}$ is **conjugate** to likelihood L if $\pi \in \mathcal{F} \Rightarrow \pi(\cdot \mid x) \in \mathcal{F}$ for every x . The **natural conjugate** is the member of $\mathcal{F}$ whose kernel in θ matches $L(\theta \mid x)$ as a function of θ .

Exponential-family recipe

If $L(\theta \mid x) = h(x) c(\theta)^n \exp(\sum_j w_j(\theta) T_j(x))$, the **natural conjugate prior** has kernel

$$\pi(\theta) \propto c(\theta)^\nu \exp\left(\sum_j w_j(\theta) \tau_j\right).$$

Update: add n to ν and $\sum_i T_j(x_i)$ to τ_j .

Binomial / Bernoulli

$x_i \mid \theta \sim \text{Bin}(m, \theta)$ (m known), prior $\theta \sim \text{Beta}(\alpha, \beta)$:

$$\theta \mid x \sim \text{Beta}\left(\alpha + \sum_i x_i, \beta + nm - \sum_i x_i\right).$$

Bernoulli: $m = 1$, so $\theta \mid x \sim \text{Beta}(\alpha + \sum x_i, \beta + n - \sum x_i)$.

Negative-Binomial / Geometric

$x_i \mid \theta \sim \text{NegBin}(r, \theta)$ ($x_i \geq 0$ failures before r th success), prior $\theta \sim \text{Beta}(\alpha, \beta)$:

$$\theta \mid x \sim \text{Beta}\left(\alpha + nr, \beta + \sum_i x_i\right).$$

Geometric: $r = 1$, so $\theta \mid x \sim \text{Beta}(\alpha + n, \beta + \sum x_i)$.

Poisson

$x_i \mid \lambda \sim \text{Pois}(\lambda)$, prior $\lambda \sim \text{Gamma}(\alpha, 1/\beta)$:

$$\lambda \mid x \sim \text{Gamma}\left(\alpha + \sum_i x_i, \frac{1}{\beta + n}\right).$$

Gamma (rate) / Exponential

$x_i \mid \lambda \sim \text{Gamma}(k, \lambda)$ (k known shape, rate λ), prior $\lambda \sim \text{Gamma}(\alpha, 1/\beta)$:

$$\lambda \mid x \sim \text{Gamma}\left(\alpha + nk, \frac{1}{\beta + \sum_i x_i}\right).$$

Exponential: $k = 1$, so $\lambda \mid x \sim \text{Gamma}(\alpha + n, 1/(\beta + \sum x_i))$.

Uniform

$x_i \mid \theta \sim U(0, \theta)$, prior $\theta \sim \text{Pareto}(\alpha, x_0)$:

$$\theta \mid x \sim \text{Pareto}(\alpha + n, \max\{x_0, \max_i x_i\}).$$

Multinomial / Categorical

$\mathbf{x}_i \mid \mathbf{p} \sim \text{Mult}(N; p_1, \dots, p_k)$ iid for $i = 1, \dots, M$, prior $\mathbf{p} \sim \text{Dir}(\alpha_1, \dots, \alpha_k)$:

$$\mathbf{p} \mid \mathbf{x} \sim \text{Dir}(\alpha_1 + m_1, \dots, \alpha_k + m_k), \quad m_j = \sum_{i=1}^M x_{ij}.$$

Categorical: $N = 1$, so m_j counts category- j observations across M draws.

Jeffreys prior

$$\pi_J(\theta) \propto \sqrt{\det I(\theta)}, \quad I(\theta) = -\mathbb{E}\left[\frac{\partial^2 \log L}{\partial \theta \partial \theta^\top}\right].$$

Invariant under reparametrisation; often improper.

Examples.

- Bernoulli: $\pi_J(\theta) \propto \theta^{-1/2}(1-\theta)^{-1/2} = \text{Beta}(1/2, 1/2)$.
- Normal (mean μ , known σ): $\pi_J(\mu) \propto 1$ (flat).
- Normal (precision τ): $\pi_J(\tau) \propto 1/\tau$.

4 Predictive distributions

Prior / posterior predictive

$$f_{X_{n+1}}(x_{n+1}) = \int \pi(x_{n+1} \mid \theta) \pi(\theta) d\theta,$$

$$f_{X_{n+1}}(x_{n+1} \mid \mathbf{X} = \mathbf{x}) = \int \pi(x_{n+1} \mid \theta) \pi(\theta \mid x) d\theta.$$

$$\mathbb{E}[X_{n+1} \mid \mathbf{X} = \mathbf{x}] = \int \mathbb{E}[X_{n+1} \mid \theta] \pi(\theta \mid x) d\theta.$$

Integrate out θ – this is the marginal for a future observation given the data.

Beta-Binomial predictive. $\theta \sim \text{Beta}(\alpha, \beta)$, $x \mid \theta \sim \text{Bin}(n, \theta)$:

$$m(x) = \binom{n}{x} \frac{B(\alpha + x, \beta + n - x)}{B(\alpha, \beta)}.$$

Post. predictive for a Bernoulli x_{n+1} : $\mathbb{P}(x_{n+1} = 1 \mid x) = \frac{\alpha+x}{\alpha+\beta+n}$ (post. mean of θ).

Gamma-Normal predictive. $\tau \sim \text{Gamma}(\alpha, 1/\beta)$, $\mu \mid \tau \sim N(m, 1/(t\tau))$, $x_i \mid \mu, \tau \sim N(\mu, 1/\tau)$:

$$X \sim m + \sqrt{\frac{\beta(1+t)}{\alpha t}} t_{2\alpha},$$

$$X_{n+1} \mid x \sim m_n + \sqrt{\frac{\beta_n(1+t_n)}{\alpha_n t_n}} t_{2\alpha_n}.$$

Same scale-family (location-scaled $t_{2\alpha}$); only the four hyperparameters update on data.

Exp / Gamma predictive. $x_i \mid \lambda \sim \text{Gamma}(k, \lambda)$ (k known shape, rate λ), prior $\lambda \sim \text{Gamma}(\alpha, 1/\beta)$:

$$f_{X_{n+1}}(x_{n+1}) = \frac{\Gamma(\alpha+k)}{\Gamma(\alpha)\Gamma(k)} \frac{\beta^\alpha x_{n+1}^{k-1}}{(\beta+x_{n+1})^{\alpha+k}},$$

$$f_{X_{n+1}}(x_{n+1} \mid \mathbf{X} = \mathbf{x}) = \frac{\Gamma(\alpha_n+k)}{\Gamma(\alpha_n)\Gamma(k)} \frac{\beta_n^{\alpha_n} x_{n+1}^{k-1}}{(\beta_n+x_{n+1})^{\alpha_n+k}}.$$

Exponential: $k = 1$, so prior pred. $\sim \text{Lomax}(\alpha, \beta)$ and posterior pred. $\sim \text{Lomax}(\alpha_n, \beta_n)$ on $(0, \infty)$.

5 One-sample testing

Posterior odds and Bayes factor

Hypotheses H_0, H_1 with prior probabilities π_0, π_1 , data x :

$$\text{prior odds} = \frac{\pi_0}{\pi_1} = \frac{\mathbb{P}(H_0)}{\mathbb{P}(H_1)},$$

posterior odds $O_n = \frac{\mathbb{P}(H_0|x)}{\mathbb{P}(H_1|x)}$,

$$O_n = \frac{\pi_0}{\pi_1} \times \underbrace{\frac{m(x|H_0)}{m(x|H_1)}}_{\text{Bayes factor } BF_{01}}.$$
Reject H_0 if $O_n < 1$ (or weigh by decision loss). $BF_{10} = 1/BF_{01}$, so evidence *against* H_0 grows with BF_{10} . **Kass–Raftery scale** for $2 \ln BF_{10}$: 0–2 weak, 2–6 positive, 6–10 strong, >10 very strong evidence vs H_0 .

Region vs region

$H_0 : \theta \in \Theta_0$ vs $H_1 : \theta \in \Theta_1$ with $\Theta_0 \cup \Theta_1$ the support.

$$\mathbb{P}(H_j \mid x) = \int_{\Theta_j} \pi(\theta \mid x) \, d\theta.$$

Evaluate via the posterior CDF over Θ_j , or Normal approximation of Beta using $\alpha_n/(\alpha_n + \beta_n)$ and its variance.

Point null $\{\theta = \theta_1\}$ **vs** $\{\theta \in \Theta_2\}$

Any continuous prior gives $\mathbb{P}(\theta = \theta_1) = 0$, so use a **mixture prior**:

$\pi(\theta) = p \, \delta_{\theta_1} + (1 - p) \, Y$, Y continuous on Θ_2 .

Then

$$\mathbb{P}(\theta = \theta_1 \mid x) \propto p \, \pi(x \mid \theta_1),$$

$$\mathbb{P}(\theta \in \Theta_2 \mid x) \propto (1 - p) \int \pi_Y(\theta) \, \pi(x \mid \theta) \, d\theta.$$

Nuisance parameters

If (θ, λ) with λ nuisance, use conditional prior $\pi(\lambda \mid \theta)$:

$$L(\theta \mid x) = \int L(\theta, \lambda \mid x) \, \pi(\lambda \mid \theta) \, d\lambda,$$

then apply the machinery above to the marginal $L(\theta \mid x)$.

Example. $\theta \sim \text{Beta}(\alpha, \beta)$, $x \sim \text{Bin}(n, \theta)$, test $H_0 : \theta \leq 0.5$. Posterior $\text{Beta}(\alpha + x, \beta + n - x)$, so $\mathbb{P}(H_0 \mid x) = F_{\text{Beta}(\alpha+x, \beta+n-x)}(0.5)$.

6 Computation

Direct Monte Carlo

If $\theta^{(1)}, \dots, \theta^{(M)} \stackrel{\text{iid}}{\sim} \pi(\theta \mid x)$,

$$\mathbb{E}[g(\theta) \mid x] \approx \tfrac{1}{M} \sum_{i=1}^M g(\theta^{(i)}).$$

SE $\sim \sigma_g/\sqrt{M}$.

Importance sampling

Trial density $h(\theta)$ covering $\text{supp}(\pi(\cdot \mid x))$; weight $\omega(\theta) = \frac{\pi(\theta)\pi(x|\theta)}{h(\theta)}$.

$$\mathbb{E}[g(\theta) \mid x] \approx \frac{\sum_i g(\theta^{(i)}) \omega(\theta^{(i)})}{\sum_i \omega(\theta^{(i)})}.$$

Self-normalised: denominator absorbs $m(x)$, so unnormalised π and $\pi(x \mid \theta)$ are fine.

Choice of trial density

- $h = \pi$ (prior): unbiased but often terrible – prior far from posterior, weights concentrate on a few draws.
- $h \propto L(\theta \mid x)$ (normalised likelihood) or a Laplace approximation to $\pi(\theta \mid x)$: usually much better.
- Warning**: if h has thinner tails than posterior, weights explode – keep h at least as heavy-tailed.

Effective sample size

$$\text{ESS} = \frac{(\sum_i \omega_i)^2}{\sum_i \omega_i^2} \in [1, M].$$

Diagnostic: ESS $\ll M$ signals weight degeneracy – a few draws dominate. Aim for $\text{ESS}/M \gtrsim 0.1$.

Example: Student- t prior + Normal likelihood. With $n = 20$,

$\bar{x} = \dots$, posterior

$$\pi(\mu \mid x) \propto (1 + \mu^2/40)^{-20.5} \exp\bigl(-\tfrac{nr}{2}(\mu - \bar{x})^2\bigr),$$

has no closed form. Sample $\mu^{(i)} \sim t_{40}$ (prior) or $N(\bar{x}, 1/(nr))$ (likelihood-based h) and reweight. The latter concentrates the weights near the posterior mode.

7 Markov Chain Monte Carlo

Recipe

If every full conditional $\pi(\theta_j \mid \theta_{-j}, x)$ is sampleable, cycle through $j = 1, \dots, k$ updating one coordinate at a time. After a burn-in of w cycles, the next M draws approximate $\pi(\theta \mid x)$.

$$\mathbb{E}[g(\theta) \mid x] \approx \tfrac{1}{M} \sum_{i=w+1}^{w+M} g(\theta^{(i)}).$$

Convergence in distribution as cycles $\rightarrow \infty$; chains are autocorrelated, so SE uses effective sample size, not M .

Dirichlet via Gibbs

$f(u, v, w) \propto u^4 v^3 w^2 (1 - u - v - w)$ on the 3-simplex.

Full conditionals (after rescaling by the residual mass):

$$u \mid v, w \sim (1 - v - w) \text{Beta}(5, 2),$$

$$v \mid u, w \sim (1 - u - w) \text{Beta}(4, 2),$$

$$w \mid u, v \sim (1 - u - v) \text{Beta}(3, 2).$$

Gamma-Normal via Gibbs

With posterior parameters $(\alpha_n, \beta_n, m_n, t_n)$ from Ch. 2:

$$\mu \mid \tau, x \sim N\bigl(m_n, \tfrac{1}{t_n \tau}\bigr),$$

$$\tau \mid \mu, x \sim \text{Gamma}(\alpha_n + \tfrac{1}{2}, 1/\beta'),$$

$$\beta' = \beta_n + \tfrac{t_n}{2}(\mu - m_n)^2.$$

Initialise $\tau^{(0)} > 0$ (e.g. $1/s^2$), then alternate. Estimate e.g. $\text{CV} = \mu\sqrt{\tau}$ via $\tfrac{1}{M} \sum \mu^{(i)}\sqrt{\tau^{(i)}}$.

8 Two-sample testing

Set-up

Independent samples $x^{(1)}, x^{(2)}$ with parameters θ_1, θ_2 and independent priors $\pi_1 \pi_2$. Joint posterior factorises:

$$\pi(\theta_1, \theta_2 \mid x) \propto \pi_1(\theta_1) L_1(\theta_1) \cdot \pi_2(\theta_2) L_2(\theta_2).$$

Independence preserved post data.

Region null, e.g. $H_0 : \theta_1 < \theta_2$

$$\mathbb{P}(H_0 \mid x) = \iint_{\theta_1 < \theta_2} \pi(\theta_1 \mid x^{(1)}) \pi(\theta_2 \mid x^{(2)}) \, d\theta_1 d\theta_2.$$

Evaluate by Monte Carlo: draw $\theta_j^{(i)} \sim \pi(\theta_j \mid x^{(j)})$ and take the fraction with $\theta_1^{(i)} < \theta_2^{(i)}$. **Beta-Beta special case**: closed form in terms of incomplete Beta.

Point null $\{\theta_1 = \theta_2\}$

Zero posterior mass under a continuous joint prior – use a **mixture**

$$\pi = p \, \delta_{\{\theta_1=\theta_2\}} \pi_{\text{common}} + (1 - p) \, \pi_1 \pi_2.$$

Under H_0 , pool both samples under common parameter θ with prior π_{common} .

$$\mathbb{P}(H_0 \mid x) \propto p \int \pi_{\text{c}}(\theta) L_1(\theta) L_2(\theta) \, d\theta.$$

Nuisance in two samples

Same marginal recipe as Ch. 5: integrate λ out of each sample's likelihood, then proceed.

Example: two proportions. $x^{(j)} \sim \text{Bin}(n_j, \theta_j)$, $\theta_j \sim \text{Beta}(\alpha_j, \beta_j)$. Posteriors are independent Beta. Test $H_0 : \theta_1 < \theta_2$ by Monte Carlo: draw $\theta_j^{(i)} \sim \text{Beta}(\alpha_j + x^{(j)}, \beta_j +$

$n_j - x^{(j)})$ for $i = 1, \dots, M$, then $\mathbb{P}(H_0 \mid x) \approx \tfrac{1}{M} \sum_i \mathbf{1}\{\theta_1^{(i)} < \theta_2^{(i)}\}$. 95% CI for $\theta_1 - \theta_2$: empirical 2.5% / 97.5% quantiles of $\{\theta_1^{(i)} - \theta_2^{(i)}\}$.